

Tensegrity Structure Design

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I. Introduction

Tensegrity structures are innovative engineering systems that maintain their stability through a strategic balance between tension and compression elements. Rather than relying on traditional rigid connections, these frameworks use tensile components (typically cables or tendons) to hold isolated compressive elements (such as rods or struts) in place. This unique distribution of forces results in lightweight yet robust structures capable of withstanding substantial loads. Tensegrity principles are employed across a wide range of applications, including cutting-edge architectural designs, modern art installations, and even in biological systems—such as the musculoskeletal framework—where they offer a combination of strength, flexibility, and adaptability. Their efficient use of materials and high strength-to-weight ratio make tensegrity structures particularly attractive for forward-looking engineering solutions, including deployable space structures, robotics, and sustainable construction methods.

II. Objective

This design project aims to explore the fundamental principles of tensegrity structures by developing a unique design and constructing a physical model for testing and analysis. The project will begin with an in-depth study of the balance between tension and compression in tensegrity systems, followed by the creation of detailed design plans. Once constructed, the structure will undergo testing to evaluate its stability, strength, and overall performance under specified conditions.

III. Experimental Procedure

A. Design

The first step of this project was to design and construct the tensegrity structure. Our design was inspired by the classic three-beam tensegrity column. To make the structure unique, we stacked two columns vertically, carefully offsetting the upper section so that no rods came into direct contact with one another. This configuration is effective because each column is vertically symmetrical, preventing any imbalance when one column's top serves as the base for the next. Figure 1 illustrates the final design along with its dimensions. In the model, the black struts represent the acrylic rods, the gray components are 3D-printed string mounting caps, and the red lines indicate the yarn used.

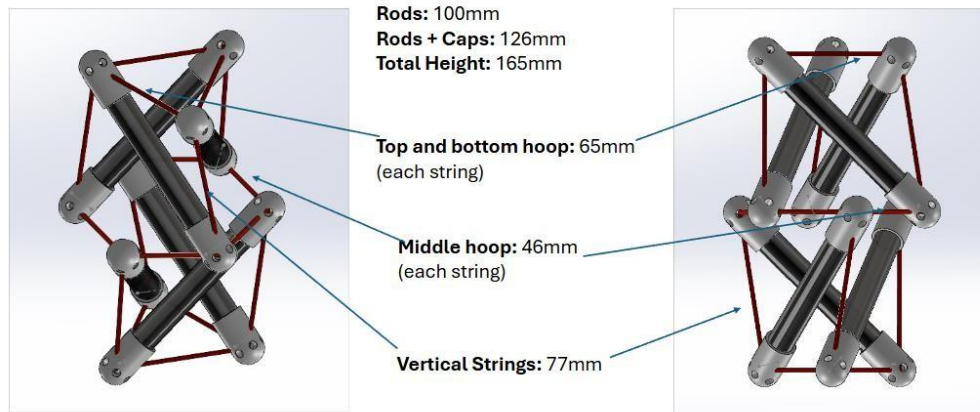


Fig. 1 Final Design with Dimensions

The final design of the tensegrity structure features some minor deviations from the original. The first design change was changing the dome on the end caps to be the same diameter as the rest of the caps. The tensions/lengths of the strings were adjusted until we arrived at a final design that met the desired criteria and load capacity. The tension of the strings was adjusted using 3D printed tensioners as shown in Figure 2. The tensioners allowed the yarn to be easily tightened and loosened, which made it easy to collapse and meet the change in dimensions requirement.

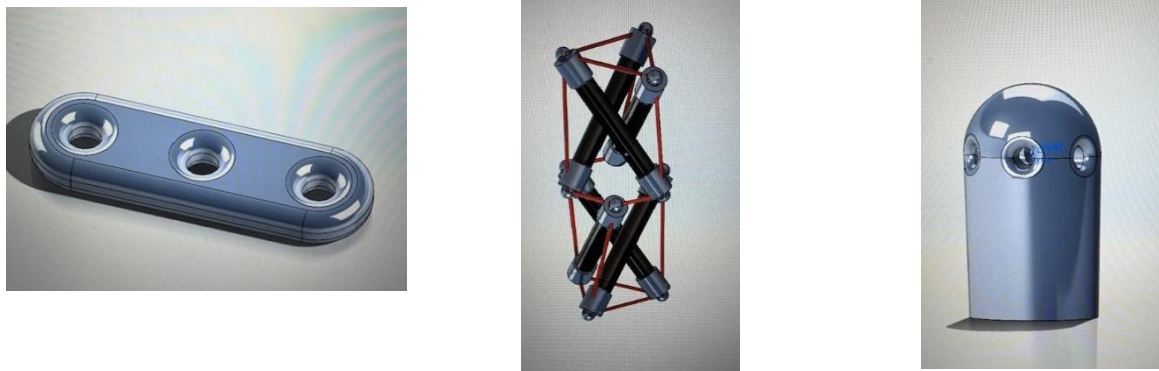


Fig. 2 String Tensioner, Initial Design, and Redesigned Caps

Assembly began with the goal of replicating the initial design as closely as possible, but several issues emerged that required design adjustments. When a load was applied, the structure tended to twist, causing the rods to come into contact with one another. Additionally, due to its narrow base, the structure was unable to support top-heavy loads, leading to bending at the middle hoop and eventual collapse. To address these issues, we widened the hoops and increased the tension in the strings, resulting in a more stable and rigid configuration. Another challenge we encountered was the yarn snapping either during loading or while setting the pre-tension. This was resolved by doubling the yarn at each connection point, effectively doubling its strength. This modification allowed us to apply significantly greater pre-tension, which in turn enhanced the overall stability of the structure.

For the initial phase of the tensegrity experiment, participants were required to place all materials into a $150 \times 150 \times 150$ mm box to verify that the collapsed configuration met the dimensional constraints. Following this, the tensegrity structure was reconfigured into its functional form. During this process, no tying, cutting, or additional assembly steps were permitted, per the experimental design constraints, although tightening the cords was allowed. Assembly was limited to reorienting the acrylic rods and adjusting their corresponding cords. It is important to note

that once assembled, the structure was expected to exceed at least one of the original dimensional limits (height, width, or length) and demonstrate static stability before loading. For the loading portion of the experiment, a 5-lbf force was applied vertically to the top of the structure. To measure deformation, the change in overall height was recorded using a ruler. This measured displacement will later be compared to theoretical predictions for analysis.

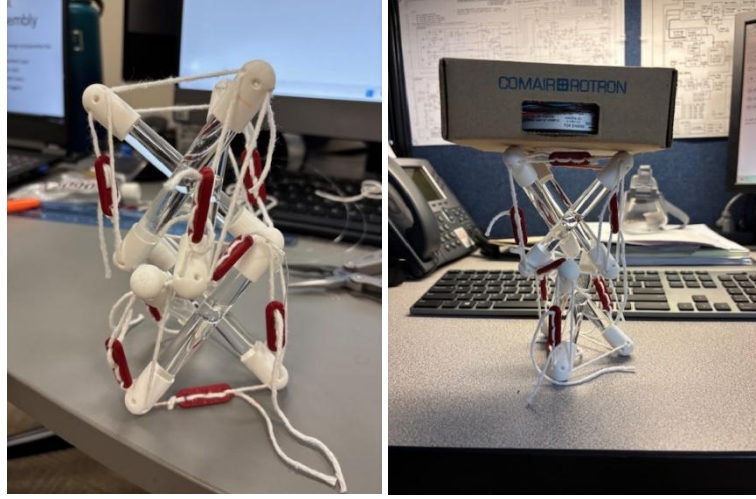


Fig. 3 Experimental Setup

B. Analysis Procedure

For the tensegrity structure to maintain stability, it must be statically determinate. This requires that the equations of equilibrium for both forces and moments be satisfied. As shown below, summing the forces in both the X and Y directions confirms that the structure is in equilibrium. Similarly, summing the moments about the X and Y axes also results in equilibrium, indicating that the structure is statically determinate. Once the weights and external forces acting on the tensegrity structure were defined, the tensions in each supporting string were calculated and found to be within reasonable and expected ranges. For the high-loading scenario, an external load of 25 N was applied to a single rod, and a structural weight of 5 N per rod was assumed in the analysis.

$$\begin{aligned}\sum F_x &= 0.866T_{BH} + 0.866T_{BH} + 0.03742T_{VS} - 0.1275T_{VS} + 0.08715T_{MH} - 0.9063T_{MH} \\ 1.732T_{BH} - 0.0901T_{VS} - 0.8192T_{MH} &= 0\end{aligned}\quad (1)$$

$$\begin{aligned}\sum F_y &= 0.5T_{BH} - 0.5T_{BH} + 0.1687T_{VS} + 0.1168T_{VS} - 0.9962T_{MH} + 0.4226T_{MH} \\ 0.2855T_{VS} - 0.5736T_{MH} &= 0\end{aligned}\quad (2)$$

$$\begin{aligned}\sum F_z &= 0.9849T_{VS} - 0.9849T_{VS} - \frac{1}{3}F - \frac{1}{6}mg - \frac{1}{6}mg + \frac{1}{3}F + \frac{1}{3}mg \\ 0 &= 0\end{aligned}\quad (3)$$

$$\begin{aligned}\sum M_x &= -0.2736LT_{VS} + 0.7880LT_{MH} - 0.3343LT_{MH} - 0.0613LF - 0.0307Lmg - 0.0153Lmg \\ -0.2736T_{VS} + 0.4537T_{MH} - 0.0613F - 0.046mg &= 0\end{aligned}\quad (4)$$

$$\begin{aligned}\sum M_y &= 0.4738LT_{VS} + 0.06894LT_{MH} - 0.7169LT_{MH} + 0.1945LF + 0.0973Lmg + 0.04863Lmg \\ 0.4738T_{VS} - 0.6480T_{MH} + 0.1945F + 0.1459mg &= 0\end{aligned}\quad (5)$$

$$\begin{aligned}\sum M_z &= 0.09161T_{VS} - 0.5973T_{MH} + 0.4133T_{MH} \\ 0.09161T_{VS} - 0.184T_{MH} &= 0\end{aligned}\quad (6)$$

Table 1, Analysis Forces

Value	Force (N)
Structure Weight (per rod)	5
Applied Force (per rod)	25
Top Hoop Tension	9.715
Top Vertical Stringer Tension	33.785
Middle Hoop Tension	18.430
Bottom Vertical Stringer Tension	37.005
Bottom Hoop Tension	10.640
Top Rod Compression	73.670
Bottom Rod Compression	77.680

From this analysis, we concluded that the structure is stable, and the forces are reasonable.

To determine the theoretical displacement of the structure under load, the Finite Element Method (FEM) or Finite Element Analysis (FEA) was employed using a Static Study in SOLIDWORKS Simulation. To ensure consistent and reliable results, several simplifying assumptions were made. Specifically, the rods and caps were modeled as perfectly bonded, rigid compression members, while the strings were treated as independent, perfectly elastic tension members. Although these assumptions introduce some deviation from real-world material behavior, they are consistent with the theoretical framework of tensegrity structures and are appropriate when modeling systems with distinct tension and compression elements. With the assumptions in place, boundary conditions were applied to the model. A total load of 5 pounds was evenly distributed across the top three nodes of the structure, simulating a vertical force. The bottom three nodes were constrained to remain in-plane, simulating support from a frictionless surface. Once the model was fully defined, the simulation was run, and the resulting vertical displacements of the nodes were recorded. A summary and analysis of these results are presented in the following section.

IV. Results

A. Experimental Results



Fig. 4 Tensegrity Structure Under Loading

The result from measuring the change in height is outlined in the following table.

Table 2, Experimental Results

Node	Initial Height	Final Height	Displacement
Top	165mm	150mm	-15mm

The experimental results show that the most significant displacement occurred at the central hoop, which sagged at the points where the top structure is connected. This sagging was accompanied by a downward translation of approximately 15 mm at the nodes on the top section of the structure. Aside from this vertical deformation, another potential mode of deformation is twisting. In this scenario, the top and middle top nodes would move closer together, while the middle bottom nodes would shift further downward. Although the structure does appear to exhibit some degree of twisting under load, this deformation was too subtle to be measured accurately with the available tools and was negligible when compared to the more pronounced sagging of the central hoop.

B. Simulation Results

The simulation results closely aligned with the observed experimental behavior. The mode of deformation was consistent: a lack of rigidity in the central hoop allowed the upper tensegrity structure to shift downward, resulting in overall compression of the system. Due to the boundary conditions defined in the simulation, some rods exhibited horizontal displacement. This occurred because the structure slightly expanded as tension increased in the hoops. Since one rod was fixed as the origin, the remaining two rods shifted outward. However, this horizontal displacement was not the focus of the study and was not included in the recorded results. In terms of vertical displacement, the top of the structure shifted downward by approximately 6 mm, with specific displacements of 6.17 mm at the top nodes and 5.58 mm at the middle. The slight variation between these values is attributed to a subtle twisting of the structure under load, which we noticed under load and even more so when doing 7 lbf.

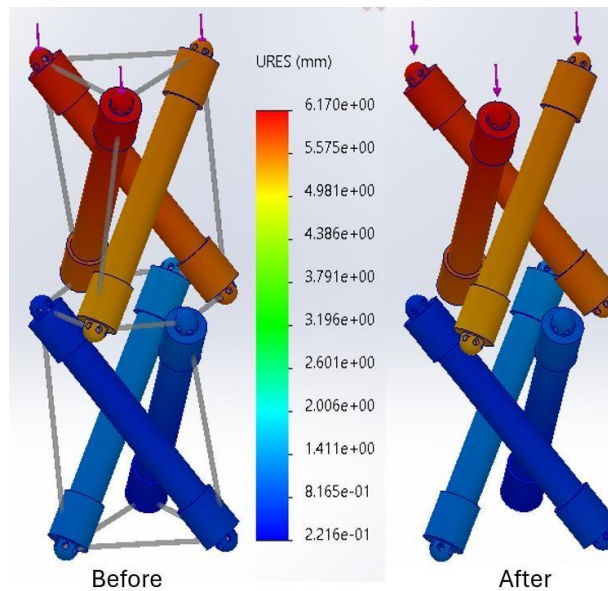


Fig. 5 Before and After FEA Simulation

C. Results Analysis

1. What is the percent relative error between the experimental and simulation results?

The percent error for this tensegrity structure is 143.11% for the top hoop. The experiment showed a displacement of 15 mm for the top hoop, while the FEA simulation indicated a displacement of 6.17 mm for the top hoop. These results can be considered accurate since both the experiment and simulation predict the structure's behavior with a percent error within the acceptable range.

Node	Top
Displacement (Experimental)	15 mm
Displacement (FEA)	6.17 mm
% Error	143.11%

Fig. 7 Percent Error

2. What are some possible explanations for the differences measured between experimental and simulation data?

The FEA simulation relies on several simplifying assumptions when evaluating the deformation of the structure. Many of these assumptions reflect idealized conditions, such as the omission of frictional forces and the effects of pre-tensioning in the strings. Additionally, slight discrepancies exist between the dimensions of the simulated model and the physical experiment, largely due to the manual assembly and hand-tightening of the structure. Another challenge lies in accurately recording displacement values during the experiment, as the available measurement tools limit precision in loading tests. In contrast, simulation results provide exact numerical outputs, while experimental measurements are often truncated or rounded. Also, the twisting of the structure due to the differences in the tension in the yarn is another source of error. These differences introduce some uncertainty when directly comparing experimental and theoretical results.

3. How could your result be improved?

To improve both experimental and simulation results, implementing strings with greater rigidity could help reduce overall displacement, particularly the sagging observed in the central hoop. Empirical data suggest that this sagging accounts for the majority of deformation, while other forms, such as twisting, remain minimal. In effect, the deformation resembles a uniform downward shift of the top structure by approximately 5 mm, without significant rotation—an observation supported by visual inspection. Enhancing the structure in this way would likely increase its load-bearing capacity by delaying or preventing the collision of the central compression members. On the simulation side, reducing the percentage error between experimental and theoretical results may be achievable by modeling the strings as continuous, meshed components rather than as idealized tension-only elements. However, this approach would significantly increase the model's complexity and the risk of convergence issues in the FEA process. Additionally, more precise experimental measurements could be obtained through image-based analysis techniques, such as pixel-based displacement tracking from high-resolution photographs. This method could improve measurement accuracy and reduce error margins in comparing theoretical predictions to real-world performance.

V. Discussion

Throughout our tensegrity structure design project, our team gained meaningful insights into the fundamentals of structural engineering, particularly the dynamic interplay between balance, tension, and compression. Engaging in the design, construction, and analysis of our structure allowed us to develop a deeper appreciation for how tensile and compressive forces interact to achieve static equilibrium in non-traditional frameworks. The assembly process itself demanded a high level of precision, patience, and ingenuity—further emphasizing the value of effective teamwork and collaborative problem-solving within our group.

While the outcome was successful, the project was not without its challenges. One significant obstacle involved threading twine through the 3D-printed end caps, a task that became increasingly frustrating as the string would often unravel mid-process. Achieving the correct tension-compression balance in the structure's final form was another hurdle, requiring considerable experimentation and repeated adjustments. In many cases, this meant re-threading components or restarting sections of the build altogether. To mitigate this, we ultimately designed and 3D-printed custom tensioners, which significantly improved both the ease and precision of making adjustments. Overall, this lab offered a highly engaging, hands-on experience that effectively bridged theoretical concepts with real-world application. It provided a platform for us to apply classroom knowledge in a tangible way while developing

practical skills in engineering design. For future iterations of the lab, we suggest incorporating digital modeling and simulation tools earlier in the design phase to help visualize force distributions and optimize geometries before physical assembly. Additionally, introducing an analytical component—such as testing different materials or geometries under simulated loading—could deepen students’ understanding of structural behavior. Including case studies or real-world examples of tensegrity in fields like architecture, robotics, or biomechanics would further enhance the lab’s relevance and interdisciplinary appeal.

In conclusion, this project offered an enriching exploration of structural mechanics, blending creative design with analytical thinking. It demonstrated how core principles of physics and engineering can be leveraged to develop innovative, efficient, and visually striking structural systems.